

Application Note: 3062



Sequence of Operations Thermostat Interface 2 PAC-US445CN-1

Introduction

The Thermostat Interface 2 (TI2) allows users to control indoor units with a third-party 24 VAC thermostat while continuing to benefit from efficiency gained through inverter technology. When set to two-stage operation, the system will begin operating at a low speed in Cool or Heat mode and adjust its capacity every 5 minutes based on how much the room temperature has increased or decreased.

See the following pages for more details on the sequence of operations for each mode.

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Required Parts

- (x1) Thermostat interface 2
 - [PAC-US445CN-1](#)
- (x1) 24 VAC Power Supply:
 - Field supplied
 - *Optional Mitsubishi part* ([VPL24-210](#))
- (x1) 24 VAC Thermostat
 - Field supplied

Important Notes

- [Click here](#) to view the TI2 Install Manual, Submittal, Compatibility Chart, and FAQ.
- All dipswitches are set to OFF out of the box.
- Be sure to configure the third-party thermostat so that it doesn't use the O/B terminals, and instead uses the Y1(Y2) and W1(W2) terminals like it would with a conventional forced-air furnace.
- The TI2 does not provide the indoor unit's status. The status can only be monitored by a central controller or MA controller connected to the indoor unit.
- Take note of the following when connecting a central controller or MA controller to the indoor unit in conjunction with a TI2.
 - Neither controller can be used to make changes to indoor unit operation. They can only be used to monitor operation and configure specific function (mode) settings. All commands will be overridden by the TI2.
 - Do not use the MA controller to change any of the following function (mode) settings.
Note: underlined settings can be changed through the TI2 dipswitches.
 - Ducted models: 01, 02, 08, 10, 11, 23, 24, 25, 27
 - Ductless models: 01, 02, 08, 10, 23, 24, 25, 27
- The indoor fan speed cannot be manually controlled via G, G1, G2, or G3 during special operations like Defrost, Pre-heat, Hot Adjust, Standby etc.
- The following steps must be taken when connecting a non-learning Nest Thermostat:
 - Connect a 2.2k 500 mW resistor to the TC and Y1 terminals of the TI2.
 - Connect a 2.2k 500 mW resistor to the C and W1 terminals of the TI2.*Note: Only required for Nest Thermostat G4CVZ models ([find your model here](#))*

All work described in this document must be performed by a licensed contractor, and must meet all national, state, and local code.

Sequence of Operations

Cool mode (Y1, Y2)

Two-stage Operation (SW2-6 OFF)

The following sequence of operations will be performed when set to two-stage operation:

Y1: When only Y1 is energized, the indoor unit will begin operating in Cool mode with a setpoint 2°F degrees lower than the current room temperature. Every 5 minutes, so long as Y1 remains energized, the indoor unit will adjust its capacity incrementally based on whether the room temperature has increased (warmer) or decreased (cooler).

- Warmer: Setpoint is lowered by 2°F degrees.
- 0-2°F cooler: Setpoint is lowered by 1°F degree.
- 3°F cooler: Setpoint remains unchanged.
- > 3°F cooler: Setpoint is raised by 1°F degree.

See Fig. 1 on [page 5](#) for the flowchart illustration of this sequence.

Y2: When Y2 is energized, the indoor unit will run at full capacity in Cool mode.

When Y1 and Y2 are both de-energized, the indoor unit will turn OFF or continue running the fan based on the SW1-1 and SW1-2 dipswitch settings, and status of the G terminal.

SW1-1 OFF: After approximately five minutes, the indoor unit will stop operating, and maintain a setpoint at least 1°F higher than the current room temperature over the next two hours. If the room temperature increases within the two-hour delay period, the indoor unit will begin cooling again to maintain the current setpoint.

During the two-hour delay period, the G1, G2, and G3 terminals can be used to manually set the fan speed to Low, Med, or High.

After two hours have passed without energizing the Y1 terminal, the indoor unit will turn OFF, and the G terminal will become active again.

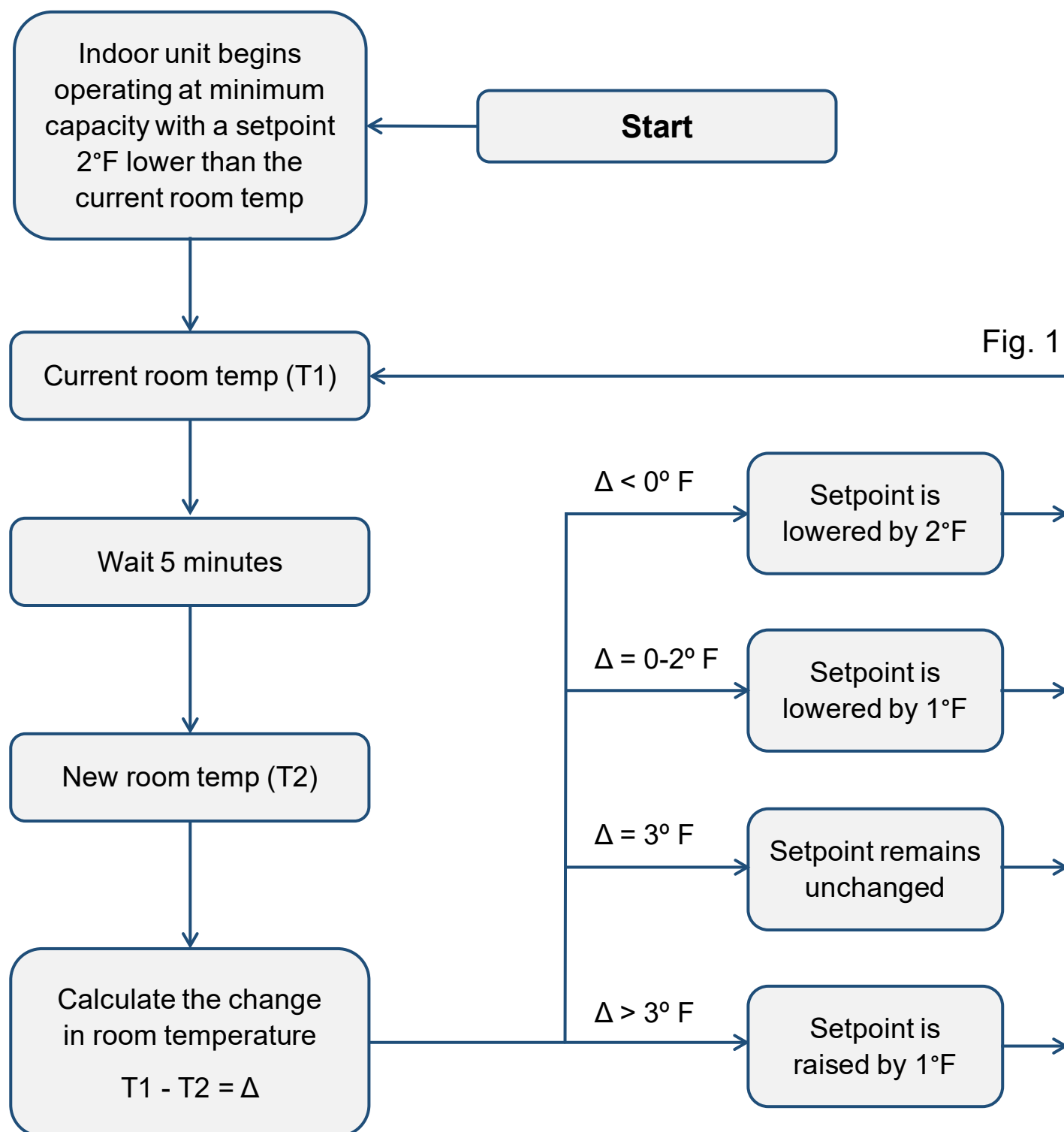
SW1-1 ON: Within a few seconds, the indoor and outdoor unit will both turn OFF, and stay off until one of the Y/W terminals is energized.

While the system is off, the G1/G2/G3 terminals will become inactive, and the G terminal will become active.

See [page 8](#) for more details on how the SW1-1 settings affect cooling operation.

Sequence of Operations

Cool mode (Y1, Y2)



Sequence of Operations

Heat mode (W1, W2)

Two-stage Operation (SW2-6 OFF)

The following sequence of operations will be performed when set to two-stage operation:

W1: When only W1 is energized, the indoor unit will begin operating in Heat mode with a setpoint 2°F degrees higher than the current room temperature. Every 5 minutes, so long as W1 remains energized, the indoor unit will adjust its capacity incrementally based on whether the room temperature has increased (warmer) or decreased (cooler).

- Cooler: Setpoint is raised by 2°F degrees.
- 0-2°F warmer: Setpoint is raised by 1°F degree.
- 3°F warmer: Setpoint remains unchanged.
- > 3°F warmer: Setpoint is lowered by 1°F degree.

See Fig. 2 on the [page 7](#) for a flowchart illustration of this sequence.

W2: When W2 is energized, the indoor unit will run at full capacity in Heat mode.

When W1 and W2 are both de-energized, the indoor unit will turn OFF or continue running the fan based on the SW1-1 and SW2-5 dipswitch settings, and status of the G terminal.

SW1-1 OFF: After approximately five minutes, the indoor unit will stop operating, and maintain a setpoint at least 1°F lower than the current room temperature over the next two hours. If the room temperature decreases within the two-hour delay period, the indoor unit will begin heating again to maintain the current setpoint.

During the two-hour delay period, the G1, G2, and G3 terminals can be used to manually set the fan speed to Low, Med, or High.

After two hours have passed without energizing the W1 terminal, the indoor unit will turn OFF, and the G terminal will become active again.

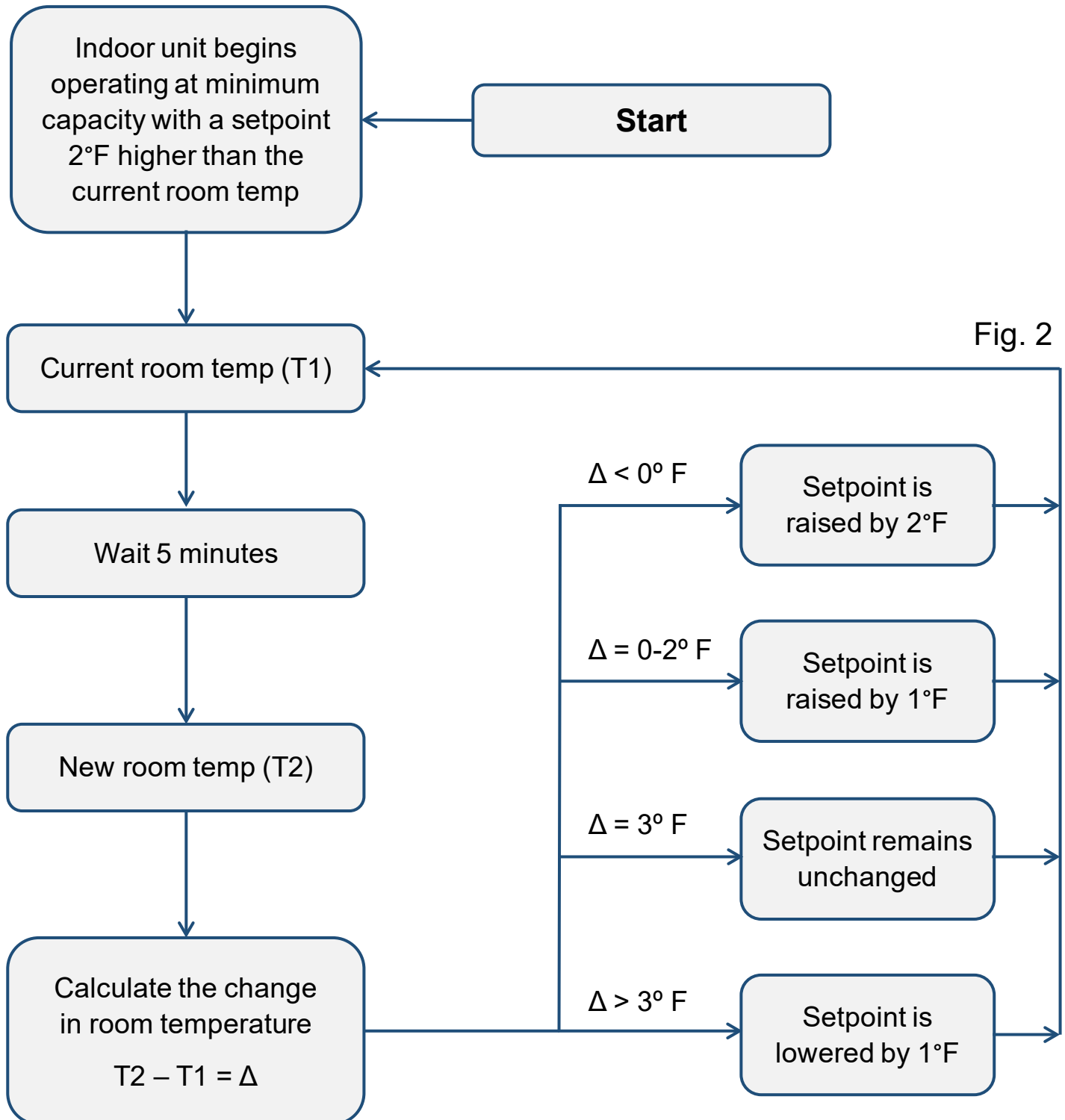
SW1-1 ON: Within a few seconds, the indoor and outdoor unit will both turn OFF, and stay off until one of the Y/W terminals is energized.

While the system is off, the G1/G2/G3 terminals will become inactive, and the G terminal will become active.

See [page 8](#) for more details on how the SW1-1 settings affect heating operation.

Sequence of Operations

Heat mode (W1, W2)



Dipswitch Settings

Thermo-OFF Time Delay (SW1-1)

The SW1-1 setting determines how the unit will operate once setpoint has been met and either of the Y or W terminals on the TI2 have been de-energized.

SW1-1 OFF (Enable Time Delay)

After approximately five minutes, the indoor and outdoor unit will stop operating. The indoor unit will then maintain a setpoint 1°F lower/higher than the current room temperature over the next two hours based on whether the system is in Heat or Cool mode. If the room temperature decreases or increases within the two-hour delay period, the indoor unit will automatically begin heating or cooling again to maintain the current setpoint, regardless of whether Y or W is energized. In addition to maintaining the setpoint, the two-hour delay also allows for operations like Defrost and Oil Recovery to be performed when necessary, which is why it's recommended to leave SW1-1 OFF whenever possible.

After two hours have passed without energizing the Y/W terminals, the indoor unit will turn OFF. The G terminal will become active, and the G1/G2/G3 terminals will become inactive.

SW1-1 ON (Disable Time Delay)

Within a few seconds, the indoor and outdoor unit will both turn OFF, and stay off until either of the Y or W terminals is energized.

While the system is off, the G1/G2/G3 terminals will become inactive, and the G terminal will become active.

Continuous Fan Operation

The following methods can be used to keep the fan operating continuously:

SW1-1 OFF: The fan can be set to run continuously in Low speed during the two-hour delay period by configuring the following dipswitches:

- Cooling: SW1-2 ON.
- Heating: SW2-5 ON.

The G1/G2/G3 terminals can adjust the fan speed during the two-hour delay period.

To keep the fan running after the two-hour delay period has expired, connect a jumper wire from the R terminal to the G terminal. This will force the indoor fan to run in High speed until either of the Y or W terminals is energized.

SW1-1 ON: Connect a jumper wire from the R terminal to the G terminal. This will force the indoor fan to run in High speed whenever the Y or W terminals is de-energized.

Dipswitch Settings

Thermo-ON/OFF (SW1-2, SW2-5)

The SW1-2 and SW2-5 dipswitch settings determine whether the fan will continue running after setpoint has been met during the two-hour time delay period described on [page 4](#).

SW1-2 ON (Cool mode, Thermo-ON)

For the remainder of the two-hour delay period, the indoor fan will continue to run in Low speed after the compressor stops. Once the two-hour delay period is finished, the fan can be turned back on by energizing the G terminal.

This setting is *not* recommended for users in hot and humid climates.

SW1-2 OFF (Cool mode, Thermo-OFF)

The indoor fan and compressor will both turn OFF at the same time. Once the two-hour delay period is finished, the fan can be turned back on by energizing the G terminal.

This setting is *highly* recommended for users in hot and humid climates

SW2-5 ON (Heat mode, Thermo-ON)

For the remainder of the two-hour delay period, the indoor fan will continue to run in Low speed after the compressor stops. Once the two-hour delay period is finished, the fan can be turned back on by energizing the G terminal.

This setting is *not* recommended when connected to an MXZ outdoor unit.

SW2-5 OFF (Heat mode, Thermo-OFF)

The indoor fan and compressor will both turn OFF at the same time. Once the two-hour delay period is finished, the fan can be turned back on by energizing the G terminal.

This setting is *highly* recommended when connected to an MXZ outdoor unit.

Important Notes

- The fan will be locked in Low speed when SW1-2 or SW2-5 is turned ON.
- The G1/G2/G3 terminals can be used to manually set the fan speed to Low, Med, during the two-hour delay period.
- The SW1-2 and SW2-5 settings have no effect when the two-hour time delay is disabled (SW1-1 On).

Dipswitch Settings

Single-stage & Two-stage Operation (SW2-6)

The SW2-6 setting determines whether the system will operate at a fixed or variable capacity when the Y1, Y2, W1, or W2 terminal is energized.

SW2-6 OFF (Two-stage, Variable Capacity)

When Y1 or W1 is energized, the system will continually increase and decrease its capacity based on the parameters described on pages 4-7.

When Y2 or W2 is energized, the system will operate at 100% capacity.

SW2-6 ON (Single-stage, Fixed Capacity)

When Y1 or W1 is energized, the system will operate at 100% capacity. The Y2 and W2 terminals will not affect system operation.

Dipswitch Settings

Default Fan Speeds (SW1-3, SW1-4)

The SW1-3 & SW1-4 settings determine how the fan will operate when set to two-stage operation (SW2-6 OFF) and either of the Y1 or W1 terminals is being energized.

SW1-3	SW1-4	Result
OFF	OFF	Custom Auto (default)
ON	OFF	Medium
OFF	ON	High
ON	ON	Auto

Custom Auto (default)

The fan speed will increase from Low to Med to High speed at a faster rate to achieve a higher level of comfort in the space at the cost of efficiency. The fan will always begin running in Low speed and eventually increase to Med-High speed as the room temperature changes. This setting is *highly* recommended for users in hot and humid climates.

Medium / High

The fan speed will run at a fixed Medium or High speed depending on the chosen setting.

Auto

The fan speed will increase at a slower rate to achieve a higher level of efficiency at the cost of comfort. This setting is *not* recommended for users in hot and humid climates.

Important Notes

- While actively cooling or heating, the G1, G2, and G3 terminals can be used to manually set the fan speed to Low, Med, or High.
- The fan will run in High speed when the Y2 or W2 terminal is energized.
- The SW1-3, and 4 settings only take effect while actively cooling or heating. If setpoint has been met, the fan will operate based on the SW1-1, SW1-2, and SW2-5 settings.

Dipswitch Settings

Static Pressure Settings (SW2-1, SW2-2, SW2-3)

The SW2-1, 2, and 3 dipswitches allow you to set the external static pressure of the indoor fan to any of the following settings on all ducted M- & P-Series models.

Static pressure settings for PAA models can be found in the PAA install manual.

Indoor Unit Models	Static Pressure Settings	Dipswitch Settings		
		SW2-1	SW2-2	SW2-3
SEZ	0.02"	ON	OFF	ON
	0.06"	ON	OFF	OFF
	0.14"	OFF	ON	OFF
	0.20"	ON	ON	OFF
PEAD	0.14"	OFF	ON	OFF
	0.20"	ON	ON	OFF
	0.28"	ON	OFF	ON
	0.40"	OFF	ON	ON
	0.60"	ON	ON	ON
SVZ PVA (Upflow) (Horizontal)	0.30"	ON	OFF	OFF
	0.50"	OFF	ON	OFF
	0.80"	ON	ON	OFF
SVZ PVA (Downflow)	0.02"	ON	OFF	ON
	0.50"	OFF	ON	ON
	0.60"	ON	ON	ON